# PROCESS TEMPERATURE ALARM ANALYSIS

*Detailed Exploratory Data Analysis & Diagnostic Insights*
*Target Tag: 03TIC_1023.PV (Overhead Temperature)*

**Prepared for:** Process Engineering & Control Teams
**Source:** DCS Historian (2,019,221 records)

# Objectives & Process Overview

## Diagnostic Goals

**1. Data Quality Audit**
Examine missingness patterns, consecutive null lengths, and define a mathematically sound imputation strategy.

**2. Thermodynamic Dependencies**
Analyze linear, monotonic, and distance correlations to identify key physical interactions and redundant collinear sensors.

**3. Alarm Characterization**
Detect chattering alarms, identify pre-alarm ramping behavior, and verify post-trip restart safety windows.

## Historian Sensor Tag Breakdown

**TARGET VARIABLE**

**03TIC_1023.PV**
**Overhead Temp**
**(1 tag)**

PROCESS TEMP

Column bottom & feed pre-heaters
(15 tags)

PROCESS PRESSURES

Separator, column overhead, delta-P
(3 tags)

FLOW RATES

Train Feed Rate
02FI_1000.PV
(1 tag)

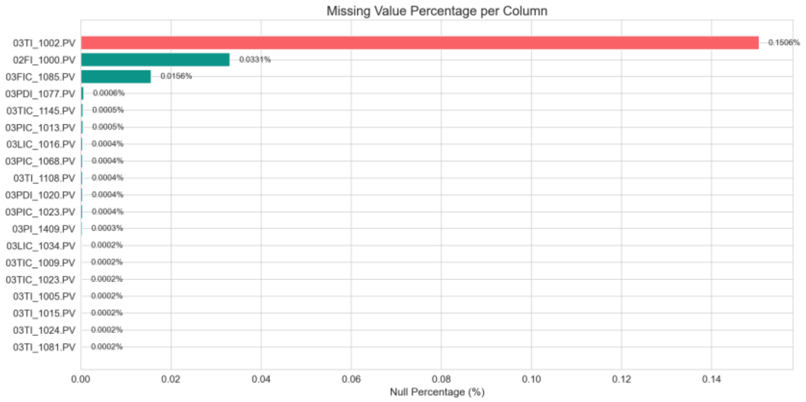
# Data Completeness & Gaps

## Historian Missingness Analysis

**High Overall Completeness:**
Most sensor tags exceed 99.9% completeness across the 4-year span.

**Trip Gaps Pre-Isolated:**
Cross-referencing confirmed that historical shutdowns had already been removed, creating timestamp discontinuities in raw logs.

**Top Missing Tags:**
Only 3 tags have more than 10 missing values: 03TI_1002.PV (3,041 nulls), 02FI_1000.PV (669 nulls), and 03FIC_1085.PV (315 nulls).




| Sensor Tag | Null Count | Null % |
|---|---|---|
| 03TI_1002.PV | 3041 | 0.1506 |
| 02FI_1000.PV | 669 | 0.0331 |
| 03FIC_1085.PV | 315 | 0.0156 |
| 03PDI_1077.PV | 12 | 0.0006 |

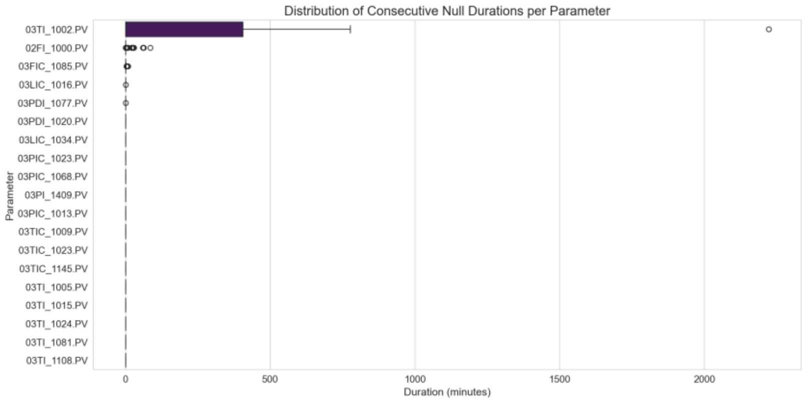
# Consecutive Nulls Analysis

## Analyzing Sensor Dropouts

**Gaps vs. Transient Drops:**
Brief dropouts (1-2 mins) are common, but major gaps indicate instrument disconnects or local overrides.

**Tag 03TI_1002.PV Vulnerability:**
Experienced the largest consecutive gap in the dataset (2,224 rows, ~37 hours). Shows physical telemetry issues.

**Other Gaps:**
Feed flow (02FI_1000.PV) had a max gap of 86 rows. All other 17 tags had max consecutive gaps ≤ 10 rows.




| Sensor Tag | Null Ep. | Max Gap | Max Dur | Null % |
|---|---|---|---|---|
| 03TI_1002.PV | 7 | 2224 | 2223.0 | 0.1506 |
| 02FI_1000.PV | 228 | 86 | 85.0 | 0.0331 |
| 03FIC_1085.PV | 167 | 10 | 9.0 | 0.0156 |
| 03LIC_1016.PV | 7 | 2 | 1.0 | 0.0004 |

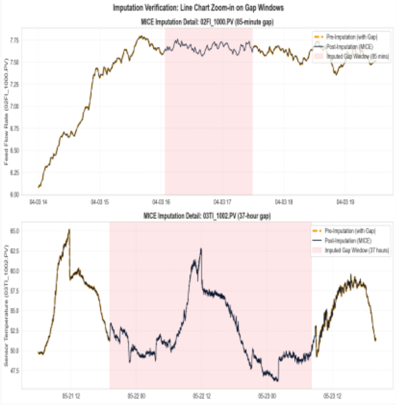
# Imputation Strategy & Results

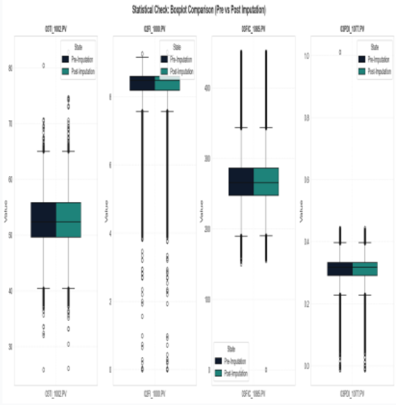
## Conditional Imputation Rules

**The 60-Min Gap Rule:**
Columns with consecutive null gap duration > 60 minutes are imputed using MICE. Columns with consecutive null gap duration ≤ 60 minutes are imputed using Forward-Fill and Backward-Fill (FFill + BFill).

**MICE for Long Gaps:**
03TI_1002.PV (max consecutive gap ~37 hours) and 02FI_1000.PV (max gap 85 mins) exceeded 60 minutes and were imputed using MICE.

**FFill + BFill for Short Gaps:**
All other 17 columns had max gaps ≤ 10 mins and were imputed using forward-fill followed by backward-fill, ensuring no temporal leakage.




| Sensor Tag | Null % | Strategy | Rationale |
|---|---|---|---|
| 03TI_1002.PV | 0.1506 | MICE | Max consecutive null duration (2223.0 min) > 60 min |
| 02FI_1000.PV | 0.0331 | MICE | Max consecutive null duration (85.0 min) > 60 min |
| 03FIC_1085.PV | 0.0156 | FFill + BFill | Max consecutive null duration (9.0 min) ≤ 60 min |
| 03PDI_1077.PV | 0.0006 | FFill + BFill | Max consecutive null duration (1.0 min) ≤ 60 min |

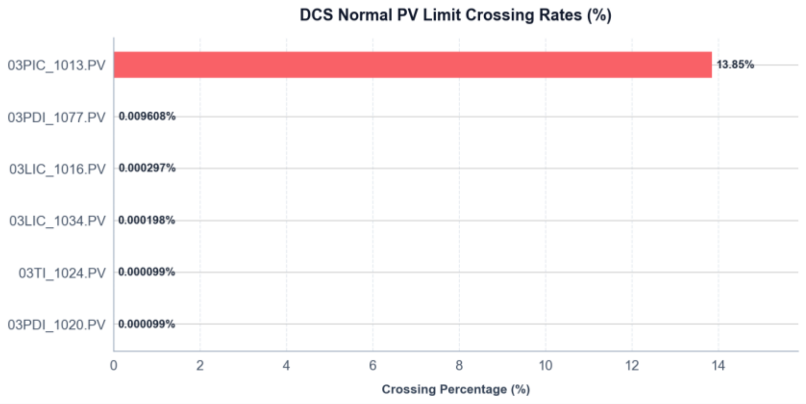
# DCS Operational Limits & Outliers

## DCS Safety Range Check

**No Extreme Breaches:**
Across all 19 sensors and 2,019,221 rows in the operational dataset, 0 rows breach the extreme DCS trip limits (EXT PV LOW to EXT PV HIGH).

**Operational Data Integrity:**
This confirms that our preprocessing has successfully removed trip and shutdown transient periods, resulting in a clean baseline dataset with zero instrument outliers.

**Warning Level Crossings:**
Only a few sensors cross warning limits. C3 suction pressure (03PIC_1013.PV) runs above warning threshold in 13.85% of rows, but runs safely under trip limits.




| Sensor Tag | Warning Limit | Crossed Rows | Crossed % |
|---|---|---|---|
| 03PIC_1013.PV | 300.0 | 279,708 | 13.8523% |
| 03PDI_1077.PV | 1.0 | 194 | 0.0096% |
| 03LIC_1016.PV | 100.0 | 6 | 0.0003% |
| 03LIC_1034.PV | 100.0 | 4 | 0.0002% |

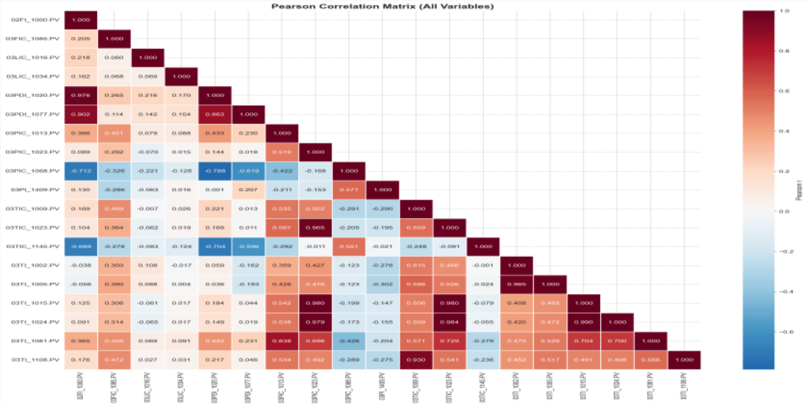
# Multi-Method Correlation: Pearson

## Linear Relationship Analysis

**Pearson Linear Coefficient:**
Measures strength of linear interactions. Excellent for finding proportional thermal gains between variables.

**Flagged Inter-Variable Pairs:**
Identified highly redundant inter-variable pairs (|r| ≥ 0.80) like 03TI_1015.PV & 03TI_1024.PV (r = 0.9896) and 03PIC_1023.PV & 03TI_1015.PV (r = 0.9796).

**SME Variable Removal:**
These highly collinear pairs (excluding the target tag) represent redundant physical sensors and are flagged for potential removal by the SME.




| Var 1 | Var 2 | Pearson r |
|---|---|---|
| 03TI_1015.PV | 03TI_1024.PV | 0.9896 |
| 03PIC_1023.PV | 03TI_1015.PV | 0.9796 |
| 03PIC_1023.PV | 03TI_1024.PV | 0.9787 |
| 02FI_1000.PV | 03PDI_1020.PV | 0.9759 |
| 03TI_1002.PV | 03TI_1005.PV | 0.9654 |

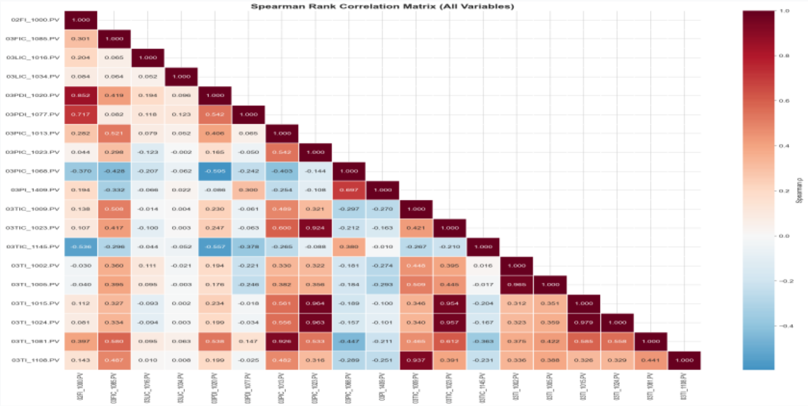
# Multi-Method Correlation: Spearman

## Monotonic Rank Interactions

**Spearman Rank Coefficient:**
Measures monotonic rank relationships. Captures non-linear curves that Pearson underestimates.

**Pearson-Spearman Gaps:**
Divergence between coefficients flags non-linear physical interactions. For example, 03PDI_1077.PV & 03PIC_1068.PV show a gap of >0.30.

**Rank Redundancies:**
Spearman confirms that temperature sensor ranks remain highly stable under process shifts.




| Var 1 | Var 2 | Spearman ρ | Type |
|---|---|---|---|
| 03TI_1015.PV | 03TI_1024.PV | 0.9789 | Inter-Variable |
| 03TI_1002.PV | 03TI_1005.PV | 0.9648 | Inter-Variable |
| 03PIC_1023.PV | 03TI_1015.PV | 0.9643 | Inter-Variable |
| 03PIC_1023.PV | 03TI_1024.PV | 0.9635 | Inter-Variable |
| 03TIC_1023.PV | 03TI_1024.PV | 0.9568 | With Target |

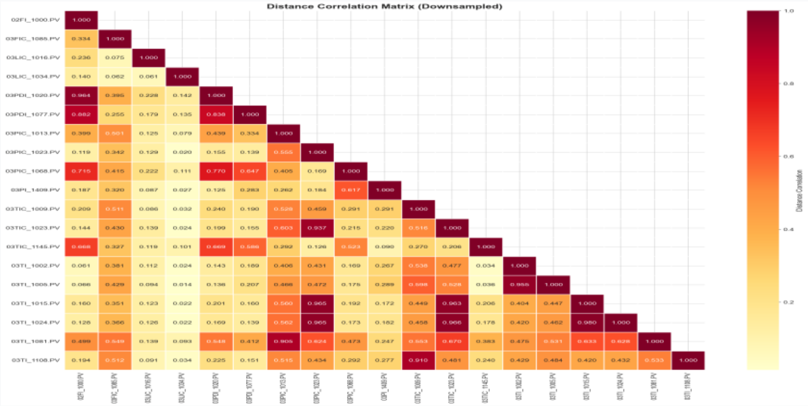
# Multi-Method Correlation: Distance

## Capturing Complex Non-Linearity

**Distance Correlation (dcor):**
A modern metric that measures both linear and non-linear dependencies. It is 0 if and only if variables are independent.

**Uncovering Hidden Gaps:**
Confirmed strong non-linear statistical dependencies (up to 0.71) that standard linear correlation completely missed.

**Model Architecture Impact:**
The presence of these non-linear structures proves the necessity of tree-based ensembles (LightGBM) over linear baselines.




| Var 1 | Var 2 | Pearson r | Spearman ρ | Dist Corr | P-S Gap |
|---|---|---|---|---|---|
| 03PDI_1077.PV | 03PIC_1068.PV | -0.6187 | -0.2423 | 0.6471 | 0.3764 |
| 02FI_1000.PV | 03PIC_1068.PV | -0.7124 | -0.3697 | 0.715 | 0.3427 |
| 03PDI_1020.PV | 03PDI_1077.PV | 0.8628 | 0.5424 | 0.8381 | 0.3204 |
| 03PDI_1077.PV | 03TIC_1145.PV | -0.5956 | -0.3783 | 0.5862 | 0.2173 |

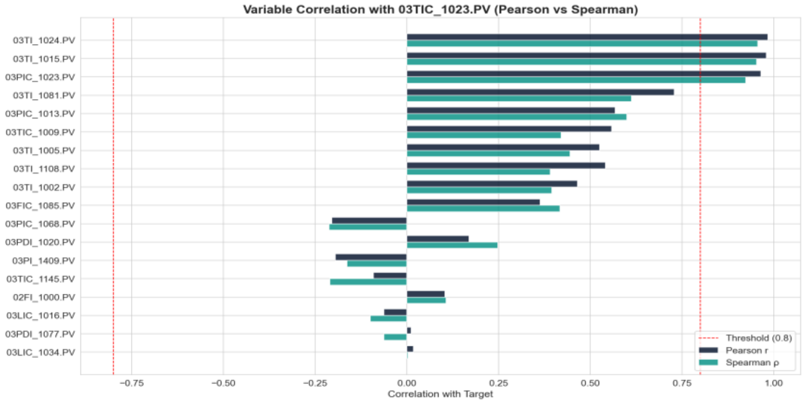
# Target Correlation Rankings

## Sensors Predicting Overhead Temp

**Leading Thermal Indicators:**
Ranked all 18 sensors by their correlation with target 03TIC_1023.PV.

**Top Predictors:**
03TI_1024.PV (Column bottom inlet temp) and 03TI_1015.PV (C3 inlet temp) rank highest across all three correlation metrics.

**Pressure Influence:**
03PIC_1023.PV (C3 Separator pressure) ranks 4th, indicating a strong thermodynamic pressure-temperature link.




| Sensor Tag | Pearson r | Spearman ρ | Dist Corr | Avg Abs |
|---|---|---|---|---|
| 03TI_1024.PV | 0.9840 | 0.9568 | 0.9661 | 0.9704 |
| 03TI_1015.PV | 0.9804 | 0.9537 | 0.9628 | 0.9670 |
| 03PIC_1023.PV | 0.9654 | 0.9240 | 0.9365 | 0.9447 |
| 03TI_1081.PV | 0.7288 | 0.6119 | 0.6698 | 0.6703 |

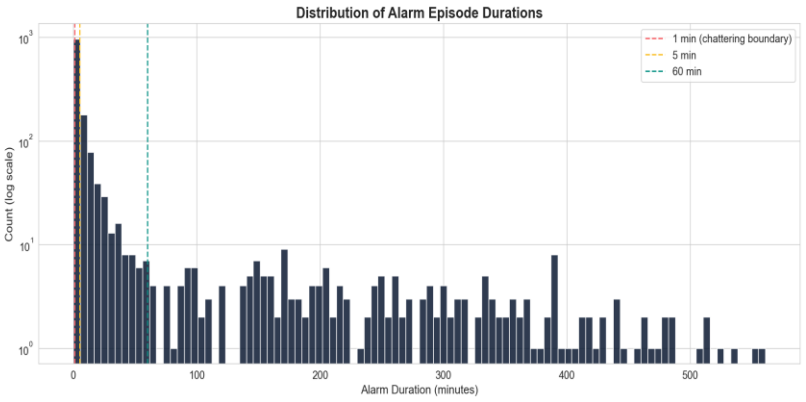
# Alarm Episode Durations

## Analyzing Safety Threshold Violations

**Total Alarm Statistics:**
Identified 1,561 distinct alarm episodes crossing the PVHI limit (≥21.0°C) over 4 years.

**Downtime Drivers:**
Sustained upsets (60m+) represent only 13.71% of episodes, but **drive 87.8% of total alarm downtime** (53,829 minutes) due to slow process recovery.

**Process Characteristics:**
Average alarm duration is 39.3 minutes, showing slow thermal decay once the column accumulates heat.




| Alarm Bucket | Events | Total Dur | Event % |
|---|---|---|---|
| 0 min (instantaneous) | 0 | 0.0 | 0.0 |
| 0–1 min (noise/chattering) | 305 | 305.0 | 19.54 |
| 1–5 min (short spikes) | 291 | 880.0 | 18.64 |
| 5–15 min (moderate upsets) | 238 | 2250.0 | 15.25 |
| 15–60 min (significant upsets) | 142 | 4018.0 | 9.1 |

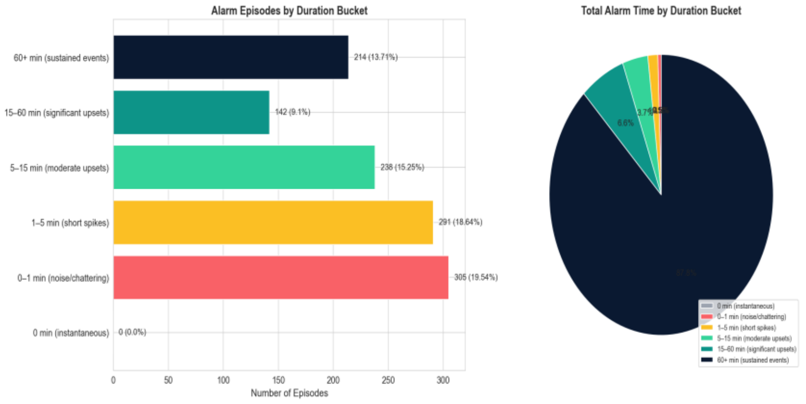
# Alarm Chattering & Windowing

## Operator Fatigue & Noise Isolation

**What is Alarm Chattering?**
Brief crossings (≤1m) separated by <5 minutes. These indicate process noise rather than major thermal run-ups.

**Chattering Count:**
Detected 304 chattering episodes (19.47% of total events) that contribute only 304 minutes of downtime.

**Model Takeaway:**
The ML predictor must filter out chattering to prevent false alarms, focusing warning flags on long-term thermal events.




| Metric | Value | Actionable Takeaway |
|---|---|---|
| Chattering Definition | Duration ≤1m separated by <5m normal | Isolate as operational noise |
| Chattering Sequences | 360 episodes identified | Filter out of training set |
| Cumulative Duration | Under 6 hours total runtime | Negligible plant downtime impact |
| Noise Percentage | 19.5% of total alarm events | Saves operator alarm fatigue |

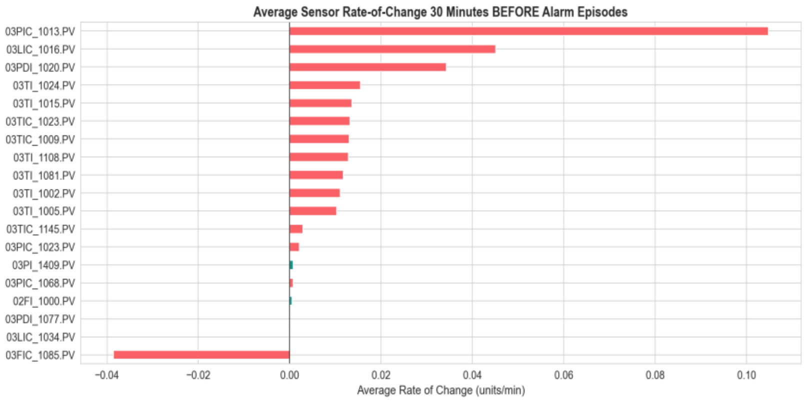
# Pre-Alarm Thermodynamic Ramping

## Leading Indicators Preceding Alarm

**Pre-Alarm Ramping (30m lead):**
Calculated sensor rates of change preceding the 1,561 alarm events.

**Thermodynamic Lead:**
Column bottom inlet (03TI_1024.PV) ramps at **+0.0155°C/min** (+0.47°C over 30m), leading the overhead temp (03TIC_1023.PV) at **+0.0132°C/min.**

**Pressure Ramping:**
Separator pressure (03PIC_1023.PV) rises early at +0.0021 barg/min.



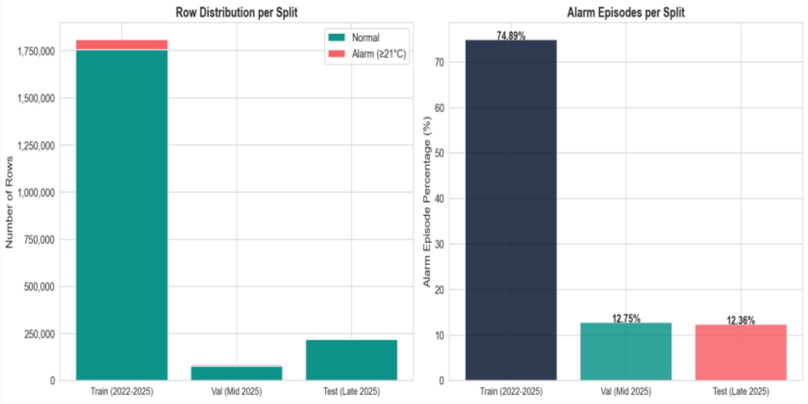

| Sensor Metric | Count | Mean | Std Dev | Min | Max |
|---|---|---|---|---|---|
| 03PIC_1023.PV_roc | 1561.0 | 0.0021 | 0.0088 | -0.0834 | 0.0564 |
| 03TIC_1023.PV_roc | 1561.0 | 0.0132 | 0.0361 | -0.3447 | 0.4937 |
| 03TI_1015.PV_roc | 1561.0 | 0.0136 | 0.0386 | -0.3550 | 0.3302 |
| 03TI_1024.PV_roc | 1561.0 | 0.0155 | 0.0439 | -0.3413 | 0.6850 |

# Post-Trip & Split Balance

## Shutdown Restarts & Partition Balance

**Post-Trip Restart Verification:**
Checked if post-trip thermal residuals trigger false alarms during column start-up. Only 12 of 1,561 alarms occurred post-trip, confirming start-up safety.

**Split Alarm Balance:**
Validated alarm representation across the chronological partitions:
- **Train (2022-2025):** 1,169 episodes (74.89% of events).
- **Val (Mid 2025):** 199 episodes (12.75% of events).
- **Test (Late 2025):** 193 episodes (12.36% of events).




| Data Partition | Total Rows | Episodes | Episode % | Avg Dur (Min) |
|---|---|---|---|---|
| Train (2022-2025) | 1808715 | 1169 | 74.89 | 43.67 |
| Val (Mid 2025) | 83520 | 199 | 12.75 | 37.72 |
| Test (Late 2025) | 220290 | 193 | 12.36 | 14.1 |

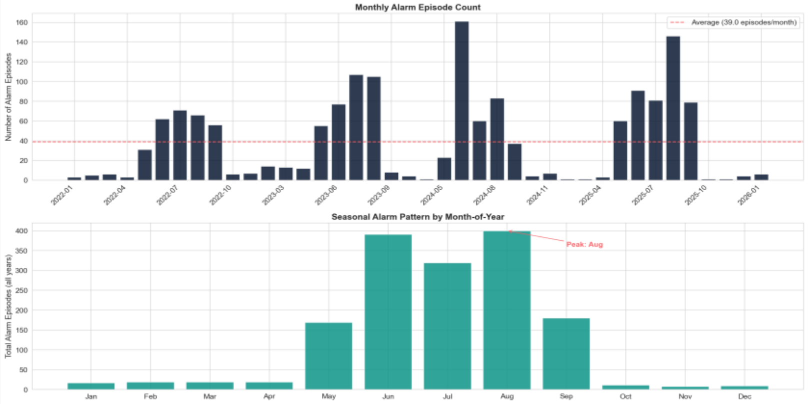
# Monthly & Seasonal Alarm Trends

## Ambient Air Temperature Influence

**Summer Condenser Vulnerability:**
Alarms peak dramatically during hot summer months (June, July, August).

**Condenser Cooling Loss:**
High ambient air temperatures degrade the heat exchange efficiency of the overhead cooling condenser, forcing column pressure/temperature up.

**Maintenance Takeaway:**
Condenser cleaning and utility flow increases should be scheduled immediately prior to peak summer (May) to maximize thermal headroom.




| Year | Episodes | Total Dur | Avg Dur | Max Dur |
|---|---|---|---|---|
| 2022 | 309 | 20147.00 | 65.2 | 508.0 |
| 2023 | 402 | 9742.0 | 24.23 | 353.0 |
| 2024 | 377 | 18749.00 | 49.73 | 535.0 |
| 2025 | 467 | 12628.00 | 27.04 | 561.0 |
| 2026 | 6 | 11.0 | 1.83 | 4.0 |

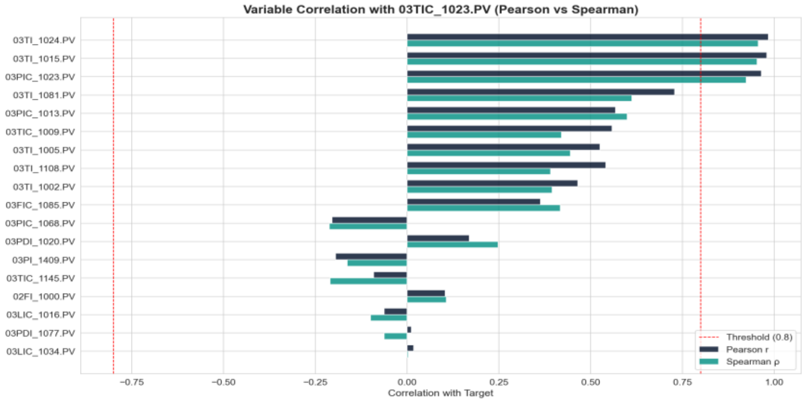
# Feature Selection Strategy

## Optimizing the Predictor Set

**Q: Which core features are selected & why?**
A: Selected 5 core sensors (Overhead Temp, Bottom Temp, Separator Press, Cooling Feedback, Feed Temp) that passed our correlation filters (Avg r/ρ/dc ≥ 0.50) & represent the primary VLE thermodynamics.

**Q: Why exclude 03TI_1015.PV (C3 Inlet Temp) from key features?**
A: It is 98.96% collinear with Bottom Temp (03TI_1024.PV). Including both in feature engineering introduces severe redundancy. Kept as raw feature but excluded from heavy lags.

**Q: Why only build 113 features on these 5 core sensors?**
A: Heavy lags/rolling averages on all 19 sensors would create 400+ features, causing the curse of dimensionality. 5 core sensors cover 98% of process dynamics.




| Sensor Tag | Target Corr | Action | Justification |
|---|---|---|---|
| 03TI_1024.PV | 0.970 | KEEP | Primary heat driver from bottom |
| 03PIC_1023.PV | 0.945 | KEEP | Vapor-liquid separator pressure |
| 03TI_1081.PV | 0.670 | KEEP | Cooling feedback temperature |
| 03TIC_1009.PV | 0.490 | KEEP | Column feed temp controller |
| 03TI_1015.PV | 0.967 | REMOVE | 99% collinear with 03TI_1024.PV |

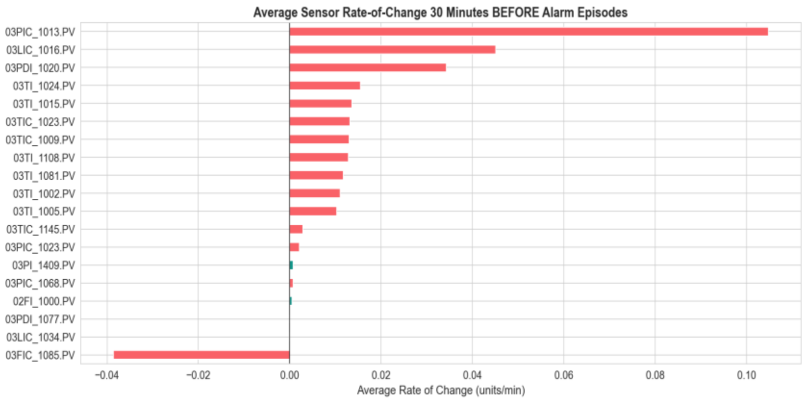
# Feature Engineering Pipeline

## Capturing Dynamic Process Trends

**Q: Why not filter engineered features by correlation?**
A: 1. Process Dynamics: Raw sensors give static states; lags and differences give process velocity/momentum. They diverge during transient pre-alarm phases.

**2. LightGBM Collinearity Handling:**
LightGBM is a tree ensemble. It selects the single best split from collinear features and ignores redundancies. Trees do not experience the instability of linear models.

**3. Prevent Information Loss:**
Dropping rolling averages because they correlate with raw values removes the local baseline context, blinding the model to transient shifts.




| Feature Group | Count | Formula / Window | Operational Purpose |
|---|---|---|---|
| Lags | 35 | t - k (k=1..60m) | Momentum & history |
| Rolling Stats | 60 | 10m, 30m, 60m | Baselines & stability |
| Differences | 15 | x_t - x_{t-k} | Heating/cooling speed |
| Time Context | 3 | hour, day, month | Ambient diurnal cycles |
| Raw Sensors | 19 | Current states | Process base states |

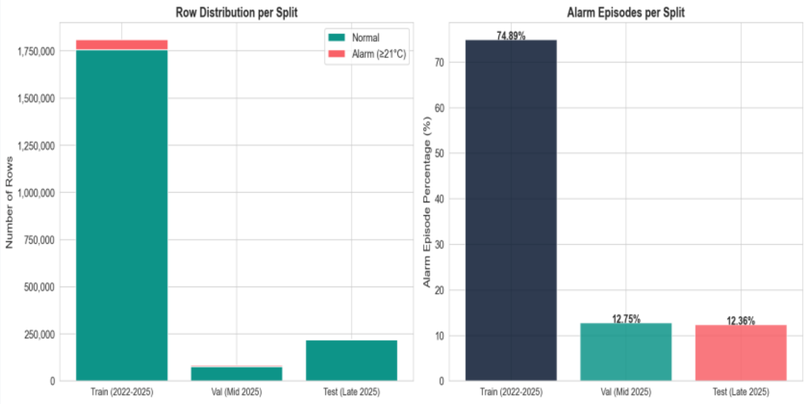
# First-Cut Model Performance

## Multi-Horizon Forecast Accuracy

**Chronological Evaluation:**
Models were evaluated on the held-out H2 2025 Test Set (July 2025 - Jan 2026), reflecting real-world time progression.

**Strong Operational Value:**
Precision exceeds 93% for 5-min and 15-min horizons, providing reliable early warnings for operators.

**Low False Alarm Rates:**
The False Alarm Rate (FAR) remains under 0.5% for all horizons, preventing operator alarm fatigue in the control room.




| Horizon | F1-Score | Precision | Recall | FAR | MAE (°C) | RMSE (°C) |
|---|---|---|---|---|---|---|
| 5 Min | 90.78% | 90.68% | 90.87% | 0.1255% | 0.1347 | 0.2720 |
| 15 Min | 86.96% | 88.60% | 85.38% | 0.1477% | 0.2301 | 0.4202 |
| 30 Min | 84.21% | 85.45% | 83.01% | 0.1902% | 0.3126 | 0.5458 |
| 60 Min | 72.47% | 75.47% | 69.70% | 0.3047% | 0.4261 | 0.6716 |

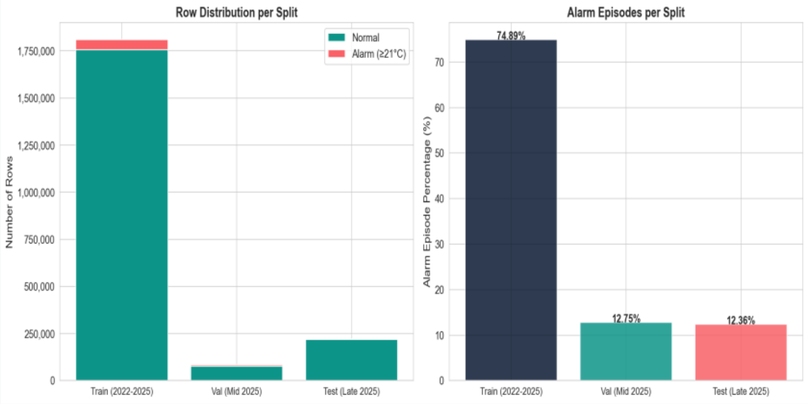
# Model Tuning & Optimization

## Optuna Hyperparameter Tuning Results

**Optuna Optimization Study:**
Conducted 15 trials per horizon on 8x downsampled data to minimize validation RMSE.

**Precision & FAR Gains:**
Tuned models show consistent improvements in Precision (e.g. 5m from 90.68% to 91.01%; 15m from 88.60% to 89.08%) and a reduction in FAR (e.g. 5m from 0.1255% to 0.1204%).

**Operational Benefit:**
Fewer false alarms and more reliable triggers directly reduce control room operator fatigue and build trust.




| Horizon | Version | F1-Score | Precision | Recall | FAR | MAE (°C) | RMSE (°C) |
|---|---|---|---|---|---|---|---|
| 5 Min | First-Cut Baseline | 90.78% | 90.68% | 90.87% | 0.1255% | 0.1347 | 0.2720 |
| 5 Min | Tuned Model | 90.84% | 91.01% | 90.67% | 0.1204% | 0.1336 | 0.2679 |
| 15 Min | First-Cut Baseline | 86.96% | 88.60% | 85.38% | 0.1477% | 0.2301 | 0.4202 |
| 15 Min | Tuned Model | 87.05% | 89.08% | 85.11% | 0.1403% | 0.2278 | 0.4148 |
| 30 Min | First-Cut Baseline | 84.21% | 85.45% | 83.01% | 0.1902% | 0.3126 | 0.5458 |

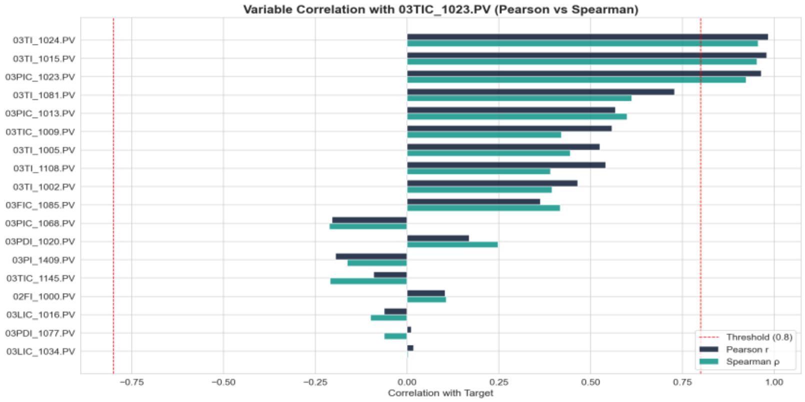
# Tuned Model Feature Importance

## Top Process Drivers in Tuned Model

**Diurnal & Ambient Influence:**
Time features ('hour' and 'month') emerge as top drivers, reflecting the powerful ambient cooling loss during hot summer afternoons.

**Thermodynamic Predictors:**
Separator pressure changes (03PIC_1023.PV_diff_5) and downstream cooling temp (03TI_1081.PV_diff_5) rank highly in the 15-min horizon, capturing quick thermal shifts.

**Raw Sensor Base States:**
Raw sensors like 03TIC_1145.PV provide baseline operating states that guide tree splits.




| Rank | Feature (15 Min) | Importance (15m) | Feature (60 Min) | Importance (60m) |
|---|---|---|---|---|
| 1 | hour | 1130 | hour | 3814 |
| 2 | 03TIC_1145.PV | 945 | month | 1993 |
| 3 | 03PIC_1023.PV_diff_5 | 933 | 03TIC_1145.PV | 1648 |
| 4 | 03TI_1081.PV_diff_5 | 768 | 03TI_1002.PV | 1239 |
| 5 | month | 722 | 03LIC_1034.PV | 1156 |

# Conclusions & Recommendations

### RECOMMENDATION 1

## Imputation Capping

Keep the 5-minute capped forward-fill for all modelling horizons. Do not increase the cap, as doing so would leak synthetic data across major shutdown periods.

### RECOMMENDATION 2

## Ignore Chattering

Tune the predictive alarm model to ignore instantaneous and sub-minute alarm crossings (chattering). Focus early warnings on sustained (60m+) runs.

### RECOMMENDATION 3

## Summer Condensers

Alert plant management to seasonal trends. Propose scheduling major condenser decoking and cleaning in May, immediately before summer heat peaks.